FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

PART TEST – II

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 231**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-C & Section-D**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01 – 03, 24 – 26, 47 – 49)*** *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*

   Partial Marks **+1** for each correct option provided no incorrect options is selected.

   Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*

2. ***Section-C (11 – 20, 34 – 43, 57 – 66)*** *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

3. ***Section-D (21 – 23, 44 – 46, 67 – 69)*** *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

PART – I (Physics), PART – II (Chemistry), PART – III (Mathematics):

(SECTION – D)

For questions **21 to 23, 44 to 46, 67 to 69.**
Numerical answer type questions with answer XXXXX. XX

If answer is 348.4 / 251.37 / 213

Correct Method :

		3	4	8	.	4	0
		2	5	1	.	3	7
		2	1	3	.	0	0

Wrong Method :

	3	4	8		.	4	
3	4	8			.		4
		3	4	8	.		4
	3		4	8	.	4	
	2		5	1	.	3	7
		2	1	3	.		
		2	1	3	.	0	
		2	1	3	.		0

Physics

PART – I

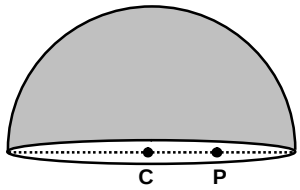
SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

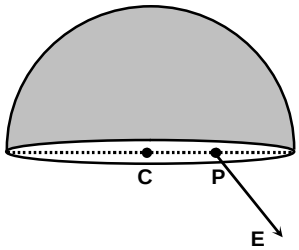
1. A non-uniform magnetic field $\vec{B} = B_0\left(1+\frac{y}{d}\right)(-\hat{k})$ is present in the region of space between y = 0 and y = d. A particle of mass m and positive charge q has velocity $\vec{v} = \frac{3qB_0d}{m}\hat{i}$ at origin O. Find the angle made by velocity of the particle with the positive x-axis when it leaves the field. (Ignore any interaction other than magnetic field)

(A) 90° (B) 30°
(C) 60° (D) 45°

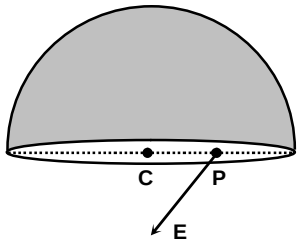
2. A thin non-conducting hemispherical shell contains a positive charge q on it, which is uniformly distributed on the shell. A point P lies on the diameter of shell as shown in figure. Then the direction of electric field at the point 'P' is



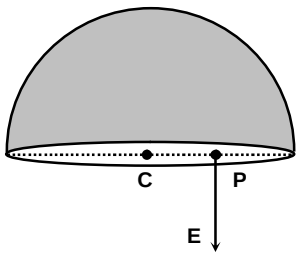
(A)



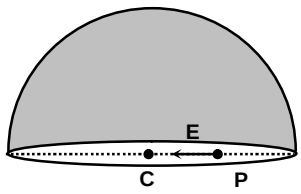
(B)



(C)

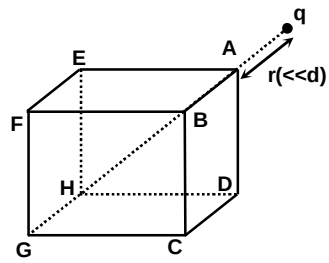


(D)



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3. A point charge q is placed on the line AH and just outside the cube (of side d) at a distance of 'r' (r << d). Total flux of electric field through the surface ABCD (ϕ_{ABCD}) is



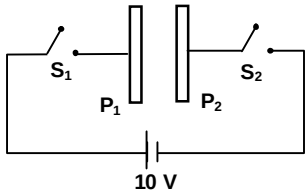
(A) $-\frac{q}{8\varepsilon_0}$ (B) $\frac{q}{16\varepsilon_0}$

(C) $\frac{q}{8\varepsilon_0}$ (D) $-\frac{q}{16\varepsilon_0}$

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

4. Two identical plates P_1 and P_2, of cross sectional area A are separated at a very small distance d, where $\frac{\varepsilon_0 A}{d} = 2$ -SI unit.



Initially charge on the plate P_1 is -10 C and charge on the plate P_2 is +30C. Now plate P_1 and P_2 are connected to positive and negative terminals of battery of potential difference 10 Volts with the help of switches S_1 and S_2 respectively. Choose the correct option(s).

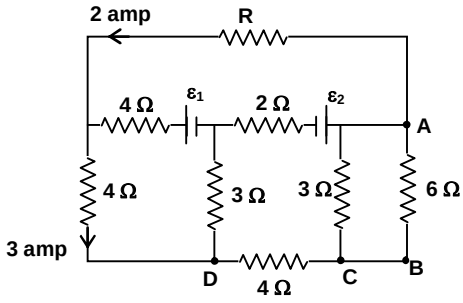
(A) The work done by the battery is 200 J.

(B) The net charge on the plate P_2 is -10C.

(C) The charge on the left side of the plate P_1 is 10C.

(D) The charge on the right side of the plate P_2 is 20C.

5. In the circuit shown, ε_1 and ε_2 are two ideal sources of unknown emfs. Some currents are shown in some branches of the circuit. Potential difference appearing across resistance 6 Ω is $V_A - V_B = 10$ V. Then



(A) The current in the CD branch is 5 amp.

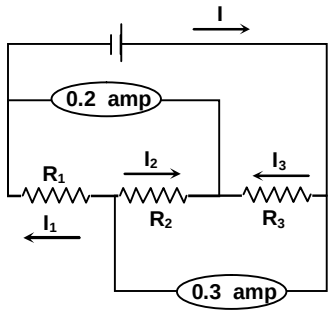
(B) The unknown emf ε_1 is 40 volts.

(C) The unknown emf ε_2 is 68 volts.

(D) The unknown resistance R is 9 Ω.

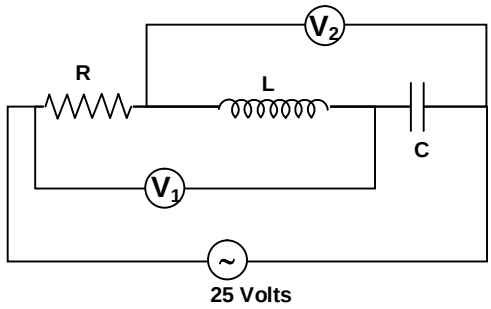
Space for Rough work

6. A circuit contains an ideal battery, three resistors and two ideal ammeters. The ammeters read 0.2 amp and 0.3 amp. After two of the resistor are switched (interchanged) the readings of ammeter do not change. Then



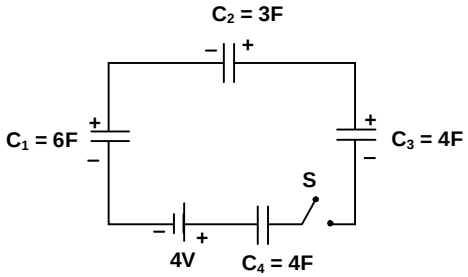
(A) If $R_2 = R_1$, $I_1 = I_2 = 0.15$ amp and $I_3 = 0.05$ amp.
(B) If $R_2 = R_1$, the battery current is, $I = 0.20$ amp.
(C) If $R_2 = R_3$, $I_2 = I_3 = 0.10$ amp and $I_1 = 0.20$ amp.
(D) If $R_2 = R_3$, the battery current is $I = 0.40$ amp.

7. In an LCR series circuit, the reading of ideal voltmeters V_1 and V_2 are $20\sqrt{2}$ volt and 15 volts respectively as shown in the figure (the given circuit is inductive in nature). Then



(A) The voltage across resistor is 20 volts.
(B) The voltage across inductor is 20 volts.
(C) The voltage across capacitor is 5 volts.
(D) The power factor of the circuit is $\frac{4}{5}$.

8. Three capacitor $C_1 = 6F$, $C_2 = 3F$ and $C_3 = 4F$ are separately charged with battery such that charge on each capacitor are 36 C, 24 C and 40 C respectively and then disconnected with the battery. After that they are connected in a circuit along with an uncharged capacitor $C_4 = 4F$ and a battery of emf = 4 volts as shown in figure. Now the switch S is closed. Choose the correct option(s).



(A) Final charge on capacitor $C_4 = 4C$.
(B) Final charge on capacitor $C_2 = 24C$.
(C) Final charge on capacitor $C_3 = 36C$.
(D) Amount of charge flow in circuit = zero.

Space for Rough work

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 9 and 10

Two moles of an ideal mono-atomic gas undergoes a thermodynamic process in which the molar heat capacity 'C' of the gas depends on absolute temperature as $C = \frac{RT}{T_0}$, where R is gas constant and T_0 is the initial temperature of the gas. (V_0 is the initial volume of the gas). Then answer the following questions.

9. The equation of process is

(A) $\frac{1}{P} = \frac{V_0 T_0^{3/2}}{4RT^{5/2}} e^{\left(\frac{T-T_0}{T_0}\right)}$ (B) $\frac{1}{P} = \frac{V_0 T_0^{3/2}}{2RT^{5/2}} e^{\left(\frac{T-T_0}{T_0}\right)}$

(C) $\frac{1}{P} = \frac{V_0 T_0^{3/2}}{4RT^{5/2}} e^{\left(\frac{T+T_0}{T_0}\right)}$ (D) $\frac{1}{P} = \frac{V_0 T_0^{3/2}}{3RT^{5/2}} e^{\left(\frac{T-T_0}{T_0}\right)}$

10. The minimum volume of the gas is

(A) $\left(\frac{2}{3}\right)^{3/2} V_0 e^{1/2}$ (B) $\left(\frac{4}{3}\right)^{3/2} V_0 e^{1/2}$

(C) $\left(\frac{5}{3}\right)^{3/2} V_0 e^{1/2}$ (D) $\left(\frac{8}{3}\right)^{3/2} V_0 e^{1/2}$

Space for Rough work

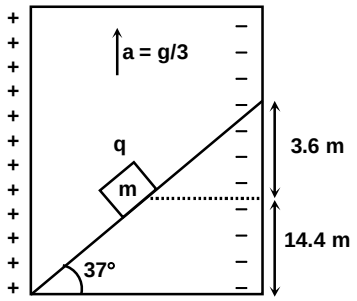
SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

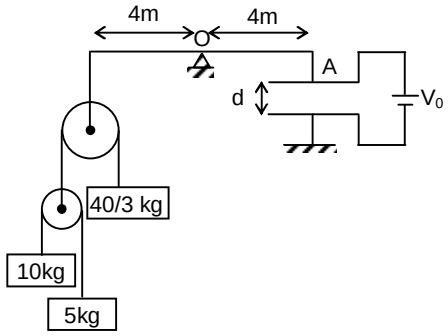
11. The electrostatic potential existing in the space is given as $V = -\left(\frac{x^3}{6\varepsilon_0} + 2\right)$ volts. Find the charge density (in coulomb/m^3) at x = 2m.

12. A hollow cube A of very small thickness at temperature T emits radiation like a black body. The cube A is kept in a room of temperature T_0(<T). The side of cube is doubled without changing the volume of material and its temperature. Find the ratio of rate of cooling of cube A in first case to that of second case.

13. A small block of mass 'm' is kept on the smooth inclined plane of angle 37°, placed in an elevator going upward with acceleration $a = \frac{g}{3}$, as shown in the figure. A horizontal electric field E perpendicular to the left and right vertical wall of the elevator exists. The charge on the block is +q. Find the time (in sec) when the block collide with the vertical wall. (take g = 10 m/s^2 and $\frac{qE}{m} = 25$ m/s^2)

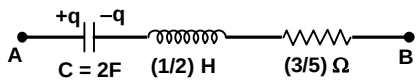


14. A capacitance balance is shown in the figure. The balance has a block pulley system attached on one side and a capacitor that has a variable gap width on the other side. Assume upper plate of the capacitor has negligible mass. When the separation between the plate is $\frac{1}{\sqrt{5}}$meter, the attractive force between the plates balances the pulley block system. Then find value of V_0 in volts (EMF of battery). (Take area of the plate = $\frac{20}{3\varepsilon_0}$m^2)

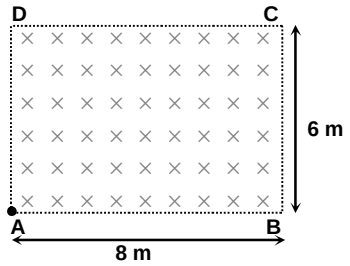


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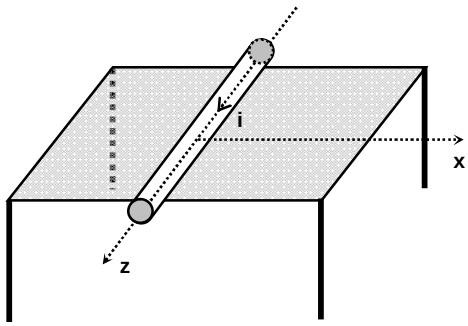
15. In the circuit shown, charge q varies with time 't' as $q = t^3 - t^2 + 4t$, where q is in coulombs and time 't' is in second. Find V_{AB} ($V_A - V_B$) in volts at time t = 1 sec.



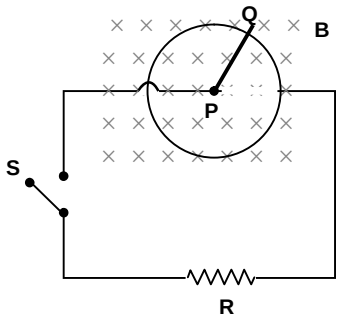
16. A charged particle of mass m = 5 gm and charge q = 12 μC enters along AB at point A in a uniform magnetic field of magnitude B = 100 Tesla existing in a rectangular region of size (8m × 6m) and into the plane of paper as shown in figure. The particle leaves the region at 'C'. The speed of the charge particle (in m/s) as it leaves the region is



17. A conducting rod of length 'ℓ' = $2\sqrt{5}$ meter and mass 'm' = 4 kg lies on the horizontal table. Coefficient of friction between the rod and the table is 'μ' $= \left(\frac{1}{2}\right)$. If the current in the conductor is 2 ampere, then find the minimum magnitude of magnetic field strength (in Tesla) such that conducting rod just starts to translate along x-axis. (take g = 10 m/sec^2) [neglect the radius of rod]

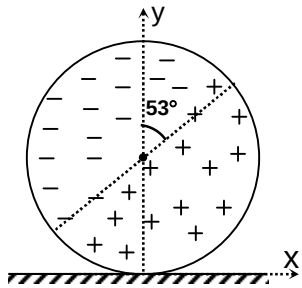


18. A resistance R = 1 kΩ connected to a conducting rod PQ that can slide on a conducting circular ring of radius 2m with center at P. A constant uniform magnetic field 10 Tesla exist as shown in figure. At t = 0, switch 'S' is closed and simultaneously an external torque is applied on the rod so that it rotates with constant angular velocity 20 rad/sec. The magnitude of power delivered by external agent is 20K Watt. Find the value of K. (Neglect the gravity, friction and self inductance of circular loop)

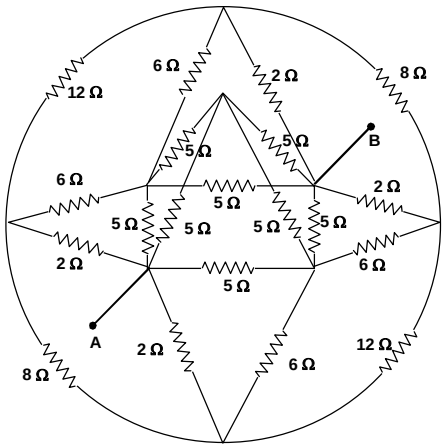


Space for Rough work

19. A non conducting solid sphere of mass m and radius R having uniform charge density $+\rho$ and $-\rho$ as shown in figure. It is then placed on a rough non conducting horizontal plane. At t = 0 a uniform electric field $\vec{E} = E_0\hat{i}$ N/C is switched on and the solid sphere starts rolling without sliding. The magnitude of frictional force at t = 0 is $\frac{2\rho\pi R^3 E_0}{K}$. Find the value of K.



20. Determine the equivalent resistance between point A and B in Ω, considering that connecting wires have no resistance.



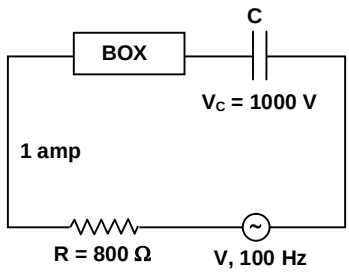
Space for Rough work

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

21. There are two vessels. Each of them contains one mole of a monoatomic ideal gas. Initial volume of the gas in each vessel is 8.3×10^{-3} m^3 at 27°C. Equal amount of heat is supplied to each vessel. In one of the vessel, the volume of the gas is doubled without change in its internal energy, where as the volume of the gas is held constant in the second vessel. The vessels are now connected to allow free mixing of the gas. Find the final temperature of the combined gas in Kelvin. (take $\ell n2 = 0.6931$)

22. A circuit shown in the figure contains a box having either a capacitor or an inductor. The power factor of the circuit is 0.8, while current lags behind the voltage. Then find the inductance/capacitance of the box in S.I. unit. (take $\pi = 3.2$) The circuit draws 1 ampere current.



23. A conductor of mass m = 5 kg and length ℓ = 4m is placed on horizontal surface and a uniform magnetic field B = 2 Tesla exist parallel to horizontal surface but perpendicular to the conductor. Suddenly, a certain amount of charge is passed through it, then it is found to jump to a height h = 5m. Then, find the amount of charge that passes through the conductor in coulomb. (take g = 10 m/s^2)

Space for Rough work

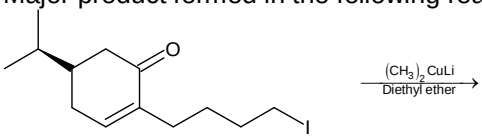
Chemistry

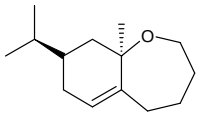
PART – II

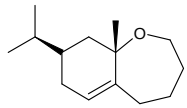
SECTION – A
(One Options Correct Type)

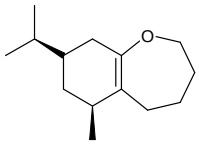
This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

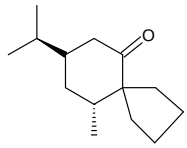
24. Major product formed in the following reaction is:



(A) 

(B) 

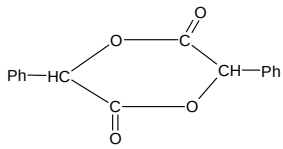
(C) 

(D) 

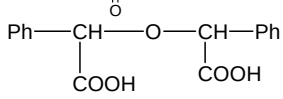
25. Identify the final product in the given reaction:

$$Ph-CH_2COOH \xrightarrow[ii)\ \bar{:}OH]{i)\ P_{red}+Br_2} X \xrightarrow{H^+,\ \Delta} Y$$

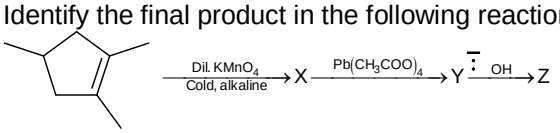
(A) $Ph-CH_2-\underset{\underset{O}{\|}}{C}-O-CH_2-Ph$

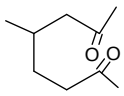
(B) 

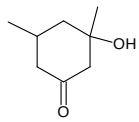
(C) $Ph-CH_2-\underset{\underset{O}{\|}}{C}-O-\underset{\underset{O}{\|}}{C}-CH_2Ph$

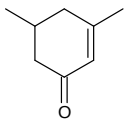
(D) 

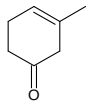
26. Identify the final product in the following reaction:



(A) 

(B) 

(C) 

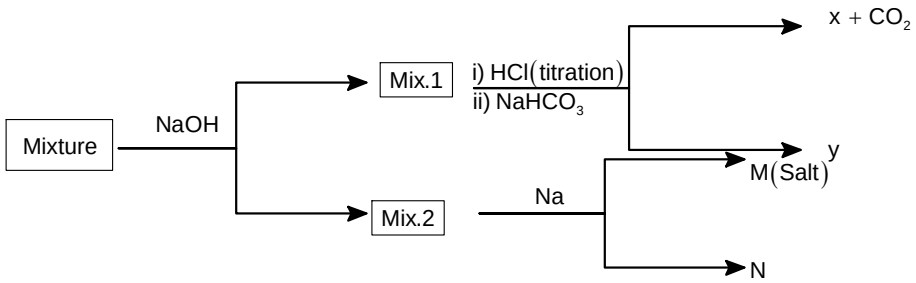
(D) 

Space for Rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

27.

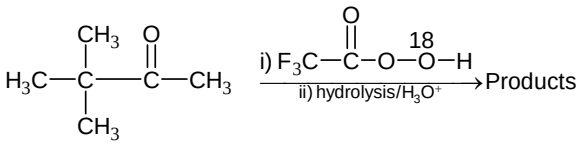


Which of the following option is correct for the above reaction?

(A) x = PhCOOH y = C_2H_5OH (B) M = PhCOONa N = CH_3OCH_3

(C) x = PhOH y = CH_3OCH_3 (D) M = C_2H_5ONa N = CH_3OCH_3

28.

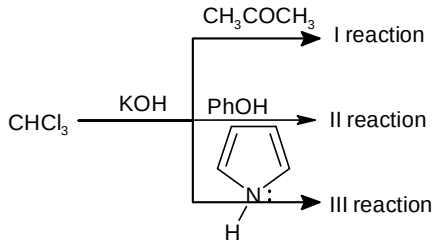


Which of the following option is/are correct?

(A) $(CH_3)_3C-\overset{18}{O}H$ (B) $CH_3CO\overset{18}{O}H$

(C) $(CH_3)_3COH$ (D) CH_3COOH

29.



in which of the following reaction carbene is intermediate

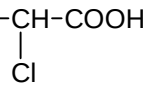
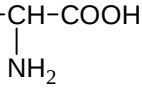
(A) I, II (B) II, III

(C) III, I (D) I, II, III

Space for Rough work

30. Which of the following reagent give same type of reaction with HCHO and PhCHO?
(A) Fehling solution (B) NH_3
(C) NaOH (D) Tollen's sol.

31. $\underset{(\alpha)}{C_3H_5N} \xrightarrow{\text{Hydrolysis}} \underset{(x)}{C_3H_6O_2} \xrightarrow[ii)\ H_3O^+]{i)\ Cl_2/P_{red}} \underset{(y)}{C_3H_5O_2Cl} \xrightarrow{NH_3} \underset{(z)}{C_3H_7NO_2}$

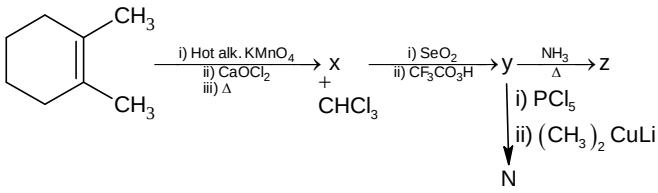
Correct option(s) is/are for the above reaction.
(A) $\alpha = C_2H_5CN$ (B) $x = C_2H_5COOH$
(C) $y =$  (D) $z =$ 

Space for Rough work

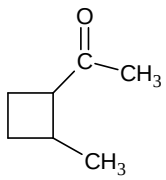
(Paragraph Type)

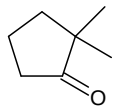
This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

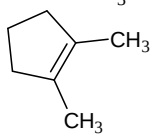
Paragraph for Question Nos. 32 and 33

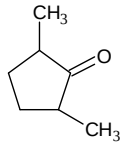


32. $N \xrightarrow[H_2O]{Mg-Hg} P \xrightarrow[Dil.\ H_2SO_4]{hot} Q$: Q is

(A) 

(B) 

(C) 

(D) 

33. $X \xrightarrow[ii)\ H_2/Ni]{i)\ KCN} R \xrightarrow[iii)\ PCl_5]{i)\ HNO_2 \quad ii)\ NH_2OH} S$

S forms a polymer. Name of polymer is:

(A) Nylon 6, 6
(B) Nylon 6
(C) Nylon 6-12
(D) Nylon 6-10

Space for Rough work

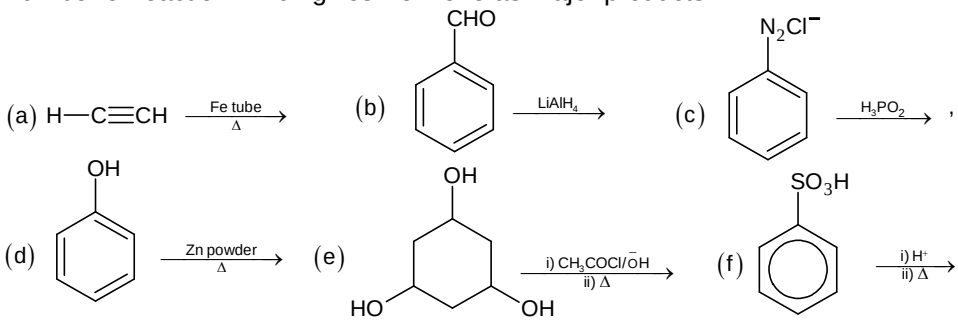
SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

34. $CH_3CH_2CHO \xrightarrow[D_2O]{DCl} Products$: Number of different enol of products is / are

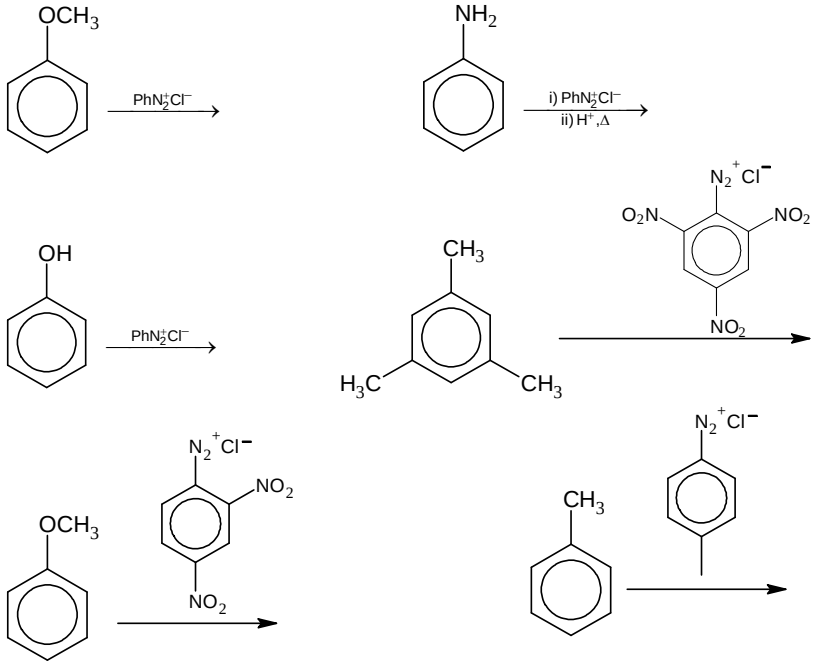
35. $PhCOOC_2H_5 + C_2H_5 - COOC_2H_5 \xrightarrow{NaNH_2}$ Number of condensated products (including stereoisomer)

36. Number of reaction which gives Benzene as major products?

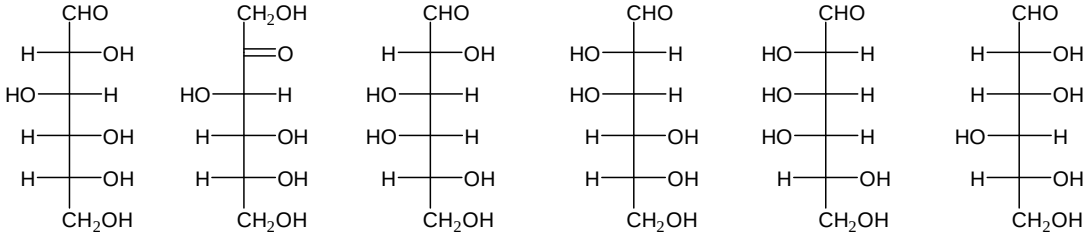


Space for Rough work

37. How many of these gives coupling reaction:



38. How many of these gives same osazone with $PhNHNH_2$:



39. How many nitrogen atoms are present in melamine monomer?

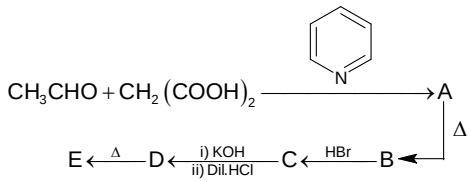
Space for Rough work

40. How many of these are thermoplastic polymers
Polyethylene, Polystyrene, Bakelite, Nylon 6,6, Terylene, Poly(methyl(methacrylate))

41. How much gm of bromine reacts with 3.15 gm of butane-2-ene (Atomic mass of bromine = 80)

42. Total number of isomers of C_4H_9Cl are

43.



Number of isomeric form of E are?

Space for Rough work

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

44. A sugar exist in two isomeric form $\alpha-D$ & $\beta-D$ with specific rotation of 110° and 30° respectively. When $\alpha-D$ form of the sugar is dissolved in water exhibits mutarotation till the equilibrium value of 90° is attained. The equilibrium constant of mutarotation of $\beta-D$ form [Assuming only α and β form in solution] is equal to:

45. A sample of 1.0 gm of organic compound was treated with NaOH and NH_3 evolved was absorbed in 70 ml 0.5 M H_2SO_4. The residual acid required 65 ml of 0.5 M NaOH solution for neutralization in Kjeldahl's method. What is the percentage of nitrogen in organic compound.

46. When 78.5 gm of CH_3COCl is treated with 29.75 gm of CH_3MgBr then what will be the mass of alcohol produces? (Given: M. wt. of Cl = 35.5, Br = 80, C = 12)

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

47. Let ABC be a right angle triangle right angle at $\angle A$, S be its circumcircle. Let S_1 be the circle touching the lines AB and AC and the circle S internally. Let S_2 be the circle touching the lines AB, AC and the circle S externally, if r_1, r_2 be the radii of the circle S_1 and S_2 respectively then area ΔABC is

(A) $\frac{r_1 r_2}{4}$ (B) $\frac{r_1 r_2}{8}$

(C) $\frac{r_1 r_2}{2}$ (D) $r_1 r_2$

48. If tangents are drawn to the parabola $y = x^2 + bx + c$ for b and c fixed real number at the points (i, y_i) for i = 1, 2, ..., 10. Let $l_1, l_2, l_3 ..... l_9$ be the point of intersection of tangents at (i, y_j) and $(i + 1, y_{i+1})$ then the least polynomial satisfying whose graph passes through all nine points

(A) $y = x^2 + bx + c$ (B) $y = x^2 + bx + c - \frac{1}{2}$

(C) $y = x^2 + bx + c - \frac{1}{4}$ (D) $y = x^2 + bx + c - \frac{1}{8}$

49. If a tangent of slope 4 of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is normal to the circle $x^2 + y^2 + 4x + 1 = 0$ then the maximum value of ab is

(A) 4 (B) 5

(C) 6 (D) 8

Space for rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

50. A(2, 0) is a point on the circle $(x + 2)^2 + (y - 3)^2 = 25$, A line through A(2, 0) making an angle of 45º with the tangent to the circle at A is drawn. Then the equations of the circles with the centres on these lines which are at a distance of $5\sqrt{2}$ units from point A and of radius 3 are
(A) $(x - 1)^2 + (y - 7)^2 = 9$ (B) $(x - 3)^2 + (y + 7)^2 = 9$
(C) $(x - 9)^2 + (y - 1)^2 = 9$ (D) $(x + 9)^2 + (y + 1)^2 = 9$

51. The normal to a curve at $P(x_1, y_1)$ meets the x-axis at G. If the distance of G from the origin is twice the abscissa of P, then the curve is a
(A) circle (B) hyperbola
(C) ellipse (D) parabola

52. The value(s) of x satisfying the equation $\tan^{-1}\left(x+\frac{2}{x}\right)-\tan^{-1}\left(x-\frac{2}{x}\right)-\tan^{-1}\left(\frac{4}{x}\right)=0$ is/are
(A) $\sqrt{3}$ (B) $-\sqrt{3}$
(C) $-\sqrt{2}$ (D) $\sqrt{2}$

53. Solution of the equation $\cos^2 x + \cos^2 2x + \cos^2 3x = 1$ is
(A) $x=(2m+1)\frac{\pi}{2}$ $(m \in I)$ (B) $x=(2n+1)\frac{\pi}{4}$ $(n \in I)$
(C) $x=(2k+1)\frac{\pi}{6}$ $(k \in I)$ (D) $x=(2\ell+1)\frac{\pi}{8}$ $(\ell \in I)$

54. Normals are drawn from the external point (h, k) to the rectangular hyperbola $xy = c^2$. If circles are drawn through the feet of these normals taken three at a time then centre of circle lies on another hyperbola whose centre and eccentricity is
(A) $\left(\frac{h}{2}, \frac{k}{2}\right)$ (B) (h, k)
(C) $\sqrt{2}$ (D) $\sqrt{2}+1$

Space for rough work

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 55 and 56

Read the following write up carefully and answer the following questions:

A variable line L intersects the parabola $y = x^2$ at points P and Q whose x-coordinate are α and β respectively with $\alpha < \beta$ the area of the figure enclosed by the segment PQ and the parabola is always equal to $\frac{4}{3}$. The variable segment PQ has its middle point as M

55. Which of the following is/are correct?
(A) $(\beta - \alpha)$ can have more than one real values (B) $(\beta - \alpha)$ can be equal to 2
(C) $(\beta - \alpha)$ can have exactly one real value (D) $\alpha = 2 + \beta$

56. Which of the following is/are correct?
(A) equations of the pair of tangents, drawn to the curve, represented by locus of M from origin are $y = 2x$ and $y = -2x$
(B) equation of pair of tangents to be curve, represented by locus of M from origin are $y = x$ and $y = -x$
(C) area of the region enclosed between the curve, represented by locus of M, and the pair of tangents drawn to it from origin is $\frac{2}{3}$ sq. units
(D) area of the region enclosed between the curve, represented by locus of M, and the pair of tangents drawn to it, from origin is $\frac{1}{3}$ sq. units

SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

57. Two tangents to an ellipse $x^2 + 2y^2 = 2$ are drawn which intersect at right angles, let ℓ_1 and ℓ_2 be the intercepts on these tangents made by the circle $x^2 + y^2 = 2$, then $\ell_1^2 + \ell_2^2$ is equal to _____

Space for rough work

58. If α, β, γ are the eccentric angles of three points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at which the normals are concurrent, then $\sin(\alpha+\beta)+\sin(\beta+\gamma)+\sin(\gamma+\alpha)$ is equal to _____

59. In a circle are inscribed a trapezoid with one side as diameter and a triangle with sides parallel to the sides of the trapezoid, then the ratio of area of trapezoid to the area and triangle is _____

60. For $x, y \in R$ with $0 < x < \frac{\pi}{2}$ such that $\frac{(\sin x)^{2y}}{(\cos x)^{\frac{y^2}{2}}}+\frac{(\cos x)^{2y}}{(\sin x)^{\frac{y^2}{2}}}=\sin 2x$, then y is _____

61. The value of cos x cos 2x·cos 3x ..... cos 999x where $x=\frac{2\pi}{1999}$ is p then $2^{1000}p$ is equal to _____

62. Normals are drawn to the parabola $y^2 = 4x$ from any point on the line y = 2, then vertices of the triangle formed by corresponding tangents lie on a fixed rectangular hyperbola $xy = -c^2$ then c^2 is _____

63. If $\frac{\tan(A+B-C)}{\tan(A-B+C)}=\frac{\tan C}{\tan B}$ then sin 2A + sin 2B + sin 2C is equal to _____

64. The line through P, perpendicular to the chord of contact of the tangents drawn from the point P to the parabola $y^2 = 16x$ touches the parabola $x^2 = 12y$, then the locus of P is $2ax + 3y + 4a^2 = 0$ then a is _____

65. Find the length of focal chord of the parabola $x^2 = 4y$ which touches the hyperbola $x^2 - 4y^2 = 1$ _____

66. Total number of solution for the equation $x^2-3\left[\sin\left(x-\frac{\pi}{6}\right)\right]=3$ is _____ (where [.] denotes the greatest integer function)

Space for rough work

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

67. Quadrilateral ABCD is inscribed in a circle with AD as diameter. If AD = 4, AB = BC =1 then perimeter of quadrilateral ABCD is _____

68. From a point (2, α) tangents are drawn to the same branch of hyperbola $\frac{x^2}{100}-\frac{y^2}{16}=1$, then the range of $\alpha \in (\ell_1, \ell_2)$, then ℓ_2 is equal to _____

69. Let ΔABC be equilateral on side AB produced, take a point P such that A lies between P and B. Consider 'a' as the length of sides of ΔABC, r_1 as the radius of incircle of ΔPAC, r_2 as the ex-radius of ΔPBC with respect to side BC, then $\frac{r_1+r_2}{\sqrt{3}\,a}$ is equal to _____

Space for rough work